Xanthan Gum Production

Modeling and Evaluation with

SuperPro Designer®

by

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This is the ReadMe file of a SuperPro Designer example that deals with process modeling, cost analysis and optimization of Xanthan Gum Production via Fermentation. The flowsheet of the process is appended to the bottom of this document. You may test-drive the model by downloading the functional evaluation edition of SuperPro Designer from the downloads page of our website (**www.intelligen.com**). The files of this example can be found in the **Examples \ Bio-Materials \ XanthanGum** folder. The default installation path of the SuperPro Designer Examples folder follows below.

C:\ Users \ Public \ Public Documents \ Intelligen \ SuperPro Designer \ v11 \ Process Library \ Examples

If you have any questions about this example and SuperPro Designer in general, please send an email message to dpetrides@intelligen.com

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Introduction

This example analyzes the production of Xanthan gum from glucose via fermentation. Xanthan gum is a water-soluble thickening and stabilizing agent, widely used in the food, healthcare, and oil industries. Xanthan gum is a hetero-polysaccharide with a primary structure consisting of repeating penta-saccharides formed by two units of glucose, two units of mannose, and one glucuronic acid, with a sequence of glucose-glucose-mannose-glucuronic acid-mannose (see Figure 1). The main chain of the polymer is identical, with cellulose consisting of D-glucose units linked with β (1→4) bonds. The difference is that every second glucose unit is linked at the O-3 position with the tri-saccharide mannose-glucuronic acid-mannose. Moreover, approximately every second terminal mannose unit is linked with pyruvic acid via a keto-group at the 4 and 6 positions, while every mannose unit near the main glucose chain is linked with an acetyl group at position 6. Each molecule consists of these stiff chains linked together, and it interacts with other molecules to form a complex. The final molecular weight varies from 2 to 20 million Dalton (Garcia-Ochoa *et al.*, 2000).

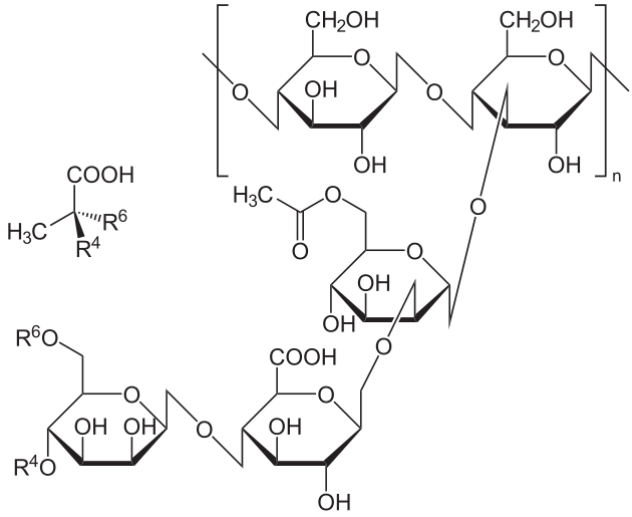


Figure 1 Structure of the xanthan gum molecule.

This complicated structure imparts xanthan gum with unique rheological properties. For example, xanthan gum substantially increases the viscosity of aqueous solutions, even at concentrations as low as 0.5% w/w, and therefore it is characterized as a thickening agent (thickener). Thickeners are used in many applications of the food and pharmaceutical industry, for instance to reduce the fat content in processed foods or to stabilize oil-in-water emulsions/suspensions, creams, milk, lotions, shampoos, toothpastes, *etc.* (Zatz and Knapp, 1984). The rheology of xanthan gum is even more interesting because its viscosity reduces with shear (Gibinski *et al.*, 2006). This shear-thinning behavior, in combination with its stability over a wide range of pH and temperatures, together with its viscosity-enhancing behavior when combined with other thickening agents, renders xanthan gum an essential additive for many products such as salad

dressings, dry mixes, syrups, beverages, dairy, as well as baked and frozen foods (Gibinski *et al.*, 2006; Saha and Bhattacharya, 2010).

Xanthan gum is produced by fermentation of bacteria of the *Xanthomonas* genus. Xanthomonas microorganisms are plant pathogens that infect a large variety of plants including cabbage, beans *etc*. The most common microorganism employed for the industrial production of xanthan gum is the *Xanthomonas Campestris* (Souw and Demain, 1979; Garcia-Ochoa *et al.*, 2000). Glucose or sucrose is typically used as the carbon source for the fermentation process. The nitrogen sources can be either organic or inorganic compounds. The xanthan gum production rate mainly depends on the concentration of carbon, whereas the biomass growth rate is *inhibited* by increasing carbon concentrations and instead primarily depends on the concentration of nitrogen (Souw and Demain, 1980; Dondo and Marques, 2002; Garcia-Ochoa *et al.*, 2000). The fermentation is aerobic with typical aeration rates in the range of 0.5-1 volumes of air per volume of liquid per minute (VVM). Adequate agitation of the thick fermentation broth is required in order to sufficiently aerate the broth and facilitate the transfer of oxygen in the fermentor. Because Xanthan gum contains acid groups in its molecular structure, a base (e.g., sodium hydroxide) is added in order to restore its pH to a neutral value in water (Garcia-Ochoa *et al.*, 2000).

Both the yield and the final quality of xanthan gum are influenced by a variety of factors such as the substrate utilized, the particular strain, the mode of operation (batch, continuous, or fed-batch), the aeration rate, the residence time, as well as the operating conditions maintained throughout the fermentation process. The pyruvic acid and acetyl groups in the molecules of xanthan gum influence its rheological properties, and therefore influence its end-use properties as well (Gibinski *et al.*, 2006; Saha and Bhattacharya, 2010). Please note that the purity and product quality requirements of the xanthan gum vary per application: the oil industry, for example, has lower quality requirements than the food and healthcare sectors.

The global xanthan gum market demand is expanding rapidly (Bixler, 1996). This trend is mainly due to the global evolution of eating habits as people in developed and developing countries are starting to prefer processed foods with particular textures. Figure 2 depicts the increasing global xanthan gum market demand in its key industries (Grand View Research). Note that the oil/gas and food/beverages sectors combined account for over 80% of the xanthan gum market volume.

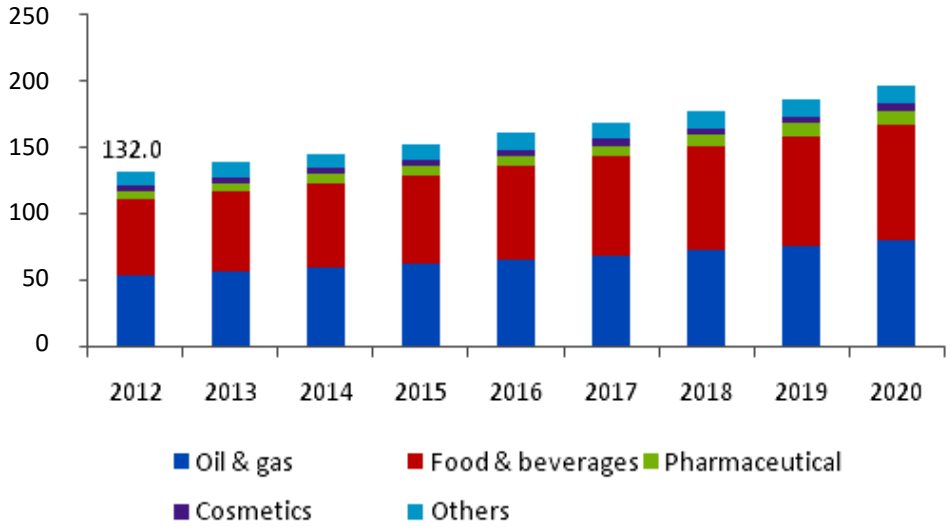


Figure 2 Global xanthan gum market volume by industry 2012-2020 (kilo-tons).

The sections that follow describe and analyze a xanthan gum production process modeled in SuperPro Designer. The development of the model was based on data available in the patent and technical literature supported by our experience with related processes and our engineering judgment.

Process Description

In order to facilitate reporting and analysis, the flowsheet was divided into four sections representing the various sets of related processing steps, namely:

- Fermentation
- Biomass Removal
- Product Purification
- Isopropanol (IPA) Recovery

Flowsheet sections in SuperPro are simply sets of related unit procedures (processing steps). The unit procedure icons of the Fermentation, Biomass Removal, Product Purification, and IPA Recovery sections are displayed in black, brown, blue, and dark green color, respectively. For information on how to specify flowsheet sections and edit their properties, please consult Chapter 8.1 of the SuperPro manual.

Fermentation

The flowsheet of the entire process is appended to the bottom of this document. The process begins with the preparation of the fermentation media, which contains glucose (~20% w/w) and nutrients such as citric acid, potassium di-hydrogen phosphate (KH_2PO), ammonium nitrate (NH_4NO_3), and various other salts in concentrations consistent with their literature values (Souw and Demain, 1980; Dondo and Marques, 2002; Garcia-Ochoa *et al.*, 2000). The media is prepared in a blending tank (V-101). For the shake of simplicity, V-101 is modeled as a continuous tank with an average residence time of 5 h, which is equivalent (in terms of equipment size) to a batch vessel operating in cycles with 5-hour durations.

The fermentation media is subsequently transferred to a storage tank (V-102) which serves as a buffer that feeds the small and large seed fermentors as well as the production fermentors. To ensure adequate media is available to feed each starting batch, the cycle time of V-102 is set to be equal to the effective cycle time of the main fermentation (16 h).

The substrate (media) solution is then passed through a continuous sterilizer (PZ-101), which sterilizes the media before it is sent into the seed and production fermentors. The fermentation water follows its own buffering and sterilization path (V-103 and PZ-102).

Note: at each scale of the fermentation (small seed, large seed, and production scale), five identical fermentors which operate in staggered mode are utilized. The specification and use of staggered mode in SuperPro models will be explained in greater detail in the **Process Scheduling** section of this document.

The inoculum for each batch is prepared in two seed fermentors operating in series. Biomass growth initially starts in a small 1.13 m^3 seed fermentor (SFR-101, 102, 103, 104, or 105) to allow for increased controllability as well as the potential to remove contaminated batches. The main inoculum preparation takes place in the large seed fermentors (SFR-201, 202, 203, 204, or 205) which each have a volume of 19 m^3. The fermentation times of the two sets of seed fermentors are 47 and 60 h, respectively. The temperature is maintained at 29°C during each seed fermentation. The conversion of glucose to xanthan gum and biomass in the seed fermentors is described by the following mass stoichiometric reaction:

1.3 Ammonium Nitrate + 2.4 Citric Acid + 100 Glucose + 3.5 KH_2PO_4 + 64 Oxygen + 0.8 Salts → 50 Biomass + 88 Carbon Dioxide + 14 Water + 20 Xanthan

The Fermentation operation in the production fermentors (FR-101, 102, 103, 104, and 105) lasts 72 h, during which oxygen is constantly supplied to the vessel by means of a compressor (G-101) at an average aeration rate of 0.5 VVM (volumes of air per volume of liquid per minute). The temperature is maintained at 31 °C, which has been shown to favor xanthan gum production over biomass growth (Souw and Demain, 1980; Dondo and Marques, 2002; Garcia-Ochoa *et al.*, 2000). The conversion of glucose to biomass and xanthan gum in the production fermentors is described by the following mass stoichiometric reaction:

1.3 Ammonium Nitrate + 2.4 Citric Acid + 100 Glucose + 3.5 KH_2PO_4 + 50 Oxygen + 0.8 Salts → 20 Biomass + 18 Carbon Dioxide + 60 Water + 60 Xanthan

The main production fermentor cycle duration is 80 h. In order to accommodate the desired production capacity, 5 identical units are operated in staggered mode, which translates into an effective cycle time of 16 h (80/5).

The seed fermentors (SFR-101 and SFR-201) and the main fermentor (FR-101) have the following operations:

- **SIP (Steam-In-Place)**. At the beginning of each batch, an SIP operation ensures sterile conditions prior to pulling in the substrate, water, and inoculum.
- **Pull-In Water**. Water pulled in from the water header FDIS-102 is fed to the seed and production fermentors to achieve a glucose concentration just below 40 g/L.
- **Pull-In Substrate**. A suitable amount of substrate (glucose) is charged.
- **Pull-In Inoculum**. The Inoculum is pulled in from the previous seed fermentor. (In the case of the small seed fermentor, the inoculum may be transferred from a container in a lab – that container is not included in this flowsheet, although SuperPro does have inoculum preparation procedures available under the Unit Procedures menu.)
- **Ferment**. Fermentation starts as soon as water, substrate and inoculum have been charged in the fermentor. A fed-batch policy for the additional supply of glucose is followed in both the seed and the production fermentors. Specifically, about 25% of the initial glucose amount is added by fed-batch feeding in the seed fermentors, while in the production fermentors the percentage is roughly 50%.
- **Pull-In NaOH**. The pH is maintained at a value around 7, which is recommended for the fermentation, by the addition of NaOH.
- **Transfer-Out**. At the end of the fermentation the vessel is emptied.
- **CIP (Clean-In-Place)**. A CIP operation takes place at the end of each batch, which includes an initial water flush, a wash with NaOH, and a final wash with water.

In order to facilitate the oxygen transfer and diffusion in the second seed and production fermentors, air is introduced under 4 bar pressure with the use of a compressor (P-12 / G-101). The temperature of the compressed air is extremely high (i.e., 228°C) and must be cooled down to 33°C before being introduced to the fermentors. The cooling takes place in cooler P-13 (HX-101), which requires a substantial cooling load of 2970 kW. This large value creates an opportunity for energy integration in the process (see the energy integration section below).

The broth of the production fermentors is transferred to a storage tank (V-104), which serves as a buffer between the Fermentation and the Biomass Removal sections. The residence time of the tank was set to 16 h. In other words, tank V-104 stores the broth of one full batch and feeds the downstream processing sections for 16 h.

Biomass Removal

After the completion of fermentation, cell mass must be deactivated and separated from the fermentation broth. Cell mass deactivation takes place in a pasteurizer (PZ-103) where it is heated with steam to 90°C and then cooled to 50°C. The inactivated cells are subsequently sent to a centrifuge (DC-101) where they are separated from the product stream. In order to recover the considerable quantity of xanthan gum that is lost during centrifugation, the solids stream (S-136) is mixed with water and centrifuged again in DC-102. The biomass recovered by the centrifuges is then dried to a final water content of 10% in a rotary drier (RDR-101), and subsequently sold as fertilizer at a price of $50/MT.

Product Purification

The product streams from the centrifuges (S-137 and S-141) are combined and sent to the purification section. There the product is precipitated out of solution with the help of salts at a concentration of 1 g/L (Smith, 1983) and isopropanol (IPA) at a concentration of 42% w/w (Smith, 1983). The precipitation takes place in a plug flow reactor (PFR-101), and the precipitation process is represented by the following "reaction" with a specified conversion of 97%:

1 Xanthan → 1 Xanthan Solids

The precipitated product stream, which has significantly lower viscosity, is sent to a centrifuge (DC-103) that separates the xanthan gum from the liquid. To minimize loss of the non-precipitated xanthan gum, a fraction of the filtrate is recycled back to mixer MX-104 at the beginning of the product purification section. The choice of the recycle percentage is a tradeoff between the 'acceptable' xanthan gum loss and the size (capacity) of the product purification section; the larger the recycled fraction, the higher the equipment and operating cost of the purification section. For the purposes of this example, an 80% recycle was specified in flow splitter P-27 and the remaining 20% is directed to the IPA recovery section. In that section, IPA is added to the precipitated xanthan gum stream (in MX-106) to a final IPA concentration of 60%. The treated stream (S-155) is concentrated to a total solids content of 50% w/w in a belt filter (BF-101) and the liquid portion is recycled back to the beginning of the product purification section. The filtered product is dried in a drum dryer (DDR-101) and the vapor stream, which contains mostly IPA, is condensed with chilled water in a condenser (HX-102) and sent to the IPA recovery section. The product stream has a flow of 620 kg/h and contains around 88% xanthan gum by weight.

IPA Recovery

IPA is used in the product purification section to facilitate the precipitation of xanthan gum and to wash the precipitated product in the belt filter. Precipitation is by far the largest IPA consumer, given that the solution needs to reach 42% IPA by weight. It is important to mention that ~18 MT/h of IPA is needed to produce ~620 kg/h of xanthan gum, a ratio of about 29:1. Consequently, recovery and reuse of IPA is critical for the economic viability of the process.

As mentioned above, 20% of the filtrate from centrifuge DC-103 (containing most of the excess salts, nutrients, and metabolites from fermentation) is sent to the IPA recovery section. The recovery takes place in a distillation column (C-101) where a distillate with 85% IPA by weight is achieved. In order to minimize energy consumption, the feed stream S-163 is preheated before entering the distillation column, initially against the distillate stream S-166, and subsequently against the bottoms stream S-167. The feed stream enters the distillation column at a temperature of around 80°C, which is close to the bubble point temperature of the mixture.

The bottoms stream, after exchanging heat with the feed of the column, is cooled to 40°C in a cooler (HX-105), and is subsequently sent to a virtual wastewater receiving storage unit. The addition and use of receiving storage units is described in a subsequent section of this document.

Finally, an IPA makeup stream with a flowrate of approximately 6 kg/h is introduced in the process, to compensate for the IPA losses.

Process Scheduling

The overall batch time for this process is approximately eight days. This is the elapsed time from the start of a given batch (i.e., the preparation of the first seed fermentor) to the end of that particular batch (the generation of the final product). However, the actual batch cycle time is only 16 hours due to the fact that the individual procedures in this process are much shorter than the overall batch time, and multiple (staggered) equipment items are used in various parts of the process.

To visualize the process schedule, click **Charts \ Equipment Occupancy \ Multiple Batches**. This will generate the Equipment Occupancy Chart (EOC). A portion of the EOC, showing 15 consecutive batches of this process, is displayed in **Error! Reference source not found.**. The EOC is a valuable tool for identifying scheduling bottlenecks (the equipment with the longest occupancy time) and performing appropriate adjustments in order to reduce the cycle time of batch portions of a process and consequently increase its throughput. For instance, Figure 3 shows that within the batch Fermentation section of this process, the production fermentors (FR-101 through 105) have no gap between consecutive batches. Therefore they are the current production bottleneck. It is also clear from this chart that operating these five fermentors in staggered mode enables the plant to initiate a new batch every 16 hours. Note that the remainder of this process operates continuously, and by default the continuous unit procedures are not displayed on the EOC (although they can be included, if desired, by right-clicking an open area of the EOC and visiting the Include/Exclude/Order Equipment dialog).

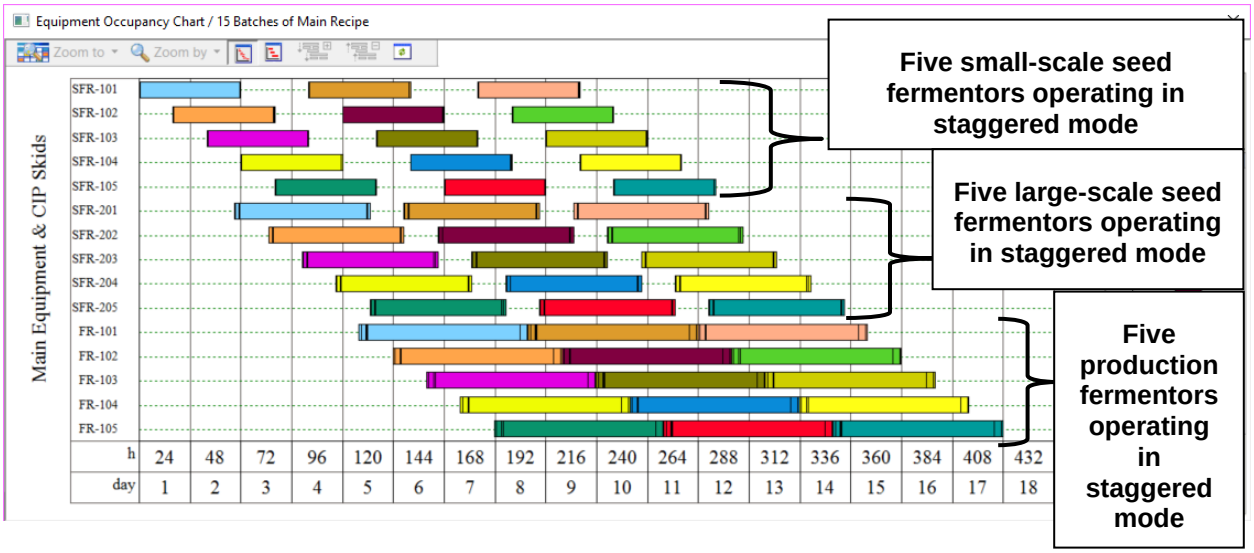


Figure 3 The Equipment Occupancy Chart for 15 batches, showing the Fermentation section.

Another view of the process schedule is provided by the Operations Gantt chart. This chart displays detailed scheduling information for one or multiple batches. The Gantt chart is generated by selecting **Charts \ Gantt Charts \ Operations GC**. Figure 4 displays the Gantt chart, illustrating the scheduling of the operations for a single batch within the two seed fermenters and the production fermentor. The brown bar at the top represents the duration of the entire batch recipe while the dark blue and cyan bars represent the durations of procedures and operations, respectively.

The Gantt chart enables users to visualize the execution of a batch process in detail. It also facilitates editing of batch recipes. Double clicking on any of its bars brings up the dialog of the corresponding entity (e.g., operation, procedure, recipe, etc.). The simulation calculations can then be redone and the chart can be updated by clicking on the refresh button of the chart ().

Furthermore, SuperPro can export its scheduling data to **MS Project** by selecting **File \ Export to MS Project XML File.** Likewise, SuperPro can export its recipe data to SchedulePro by selecting **File \ Export to SchedulePro's Recipe DB.** SchedulePro is a resource management, production planning and scheduling tool available from Intelligen. Please **c**onsult the SuperPro manual or its Help facility for information on these two exporting options.

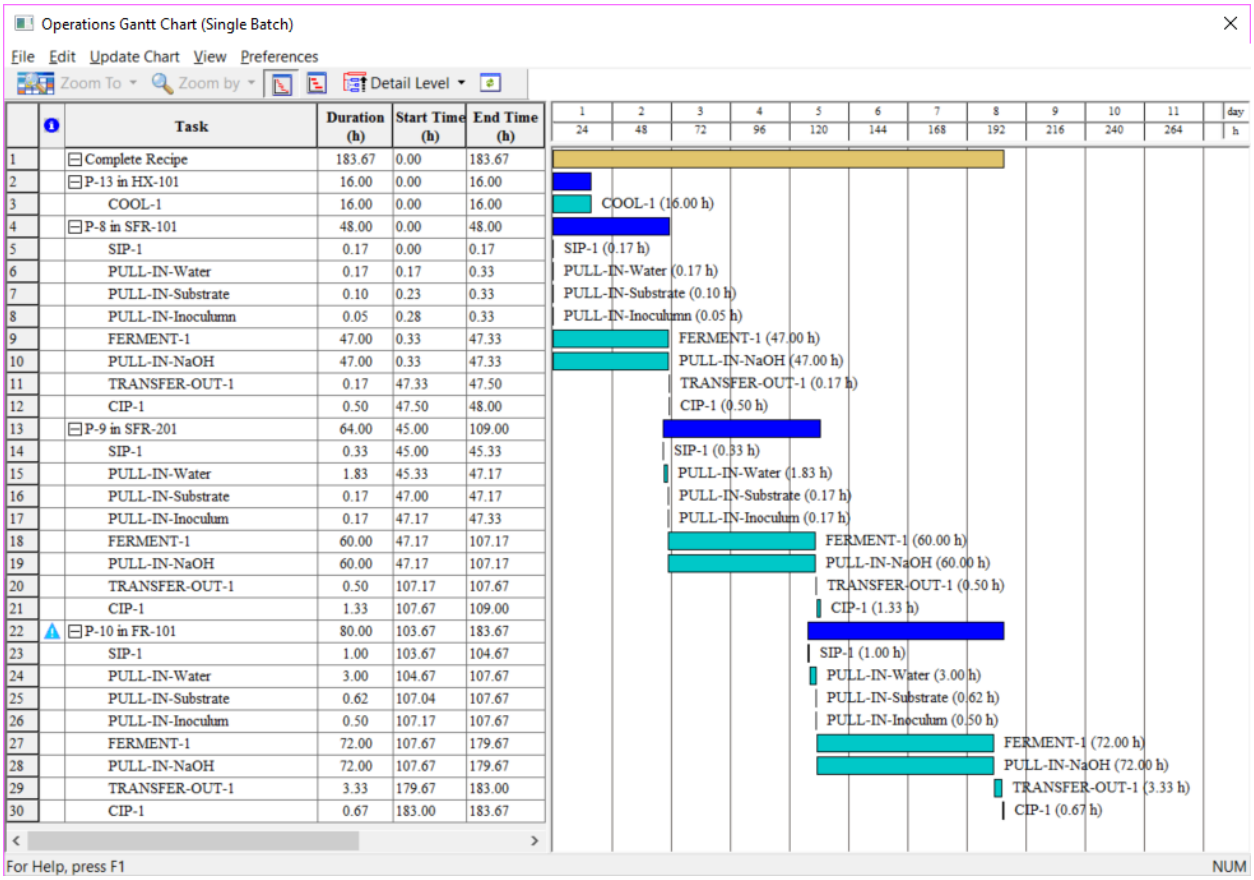


Figure 4 Operations Gantt chart.

Specifying Equipment in Staggered Mode

As mentioned in the process description section, the fermentation steps (small seed, large seed, and production fermentation) are each carried out by 5 identical fermentors that operate in staggered mode (out of phase). The Fermentation section feeds a continuous purification train.

The addition of extra equipment items that operate in staggered mode is done through the equipment data dialog of a procedure, which is accessed by right-clicking on the procedure icon and selecting **Equipment Data**. Figure 5 displays the equipment data dialog of the fermentor (FR-101). The **Stagger Mode** frame is located on the lower left corner of the dialog. To specify staggered equipment, check the **On** box of that frame and specify the number of extra pieces of equipment that are required (four extra units, in this case). The names of the extra equipment items can be edited by clicking on the **Names** button in this same frame. Notice that the total number of equipment units available to the procedure is equal to the number of units in stagger mode, plus one. When a unit procedure is set to Stagger mode, a new icon ( ) will appear by the lower left corner of that procedure's icon on the flowsheet.

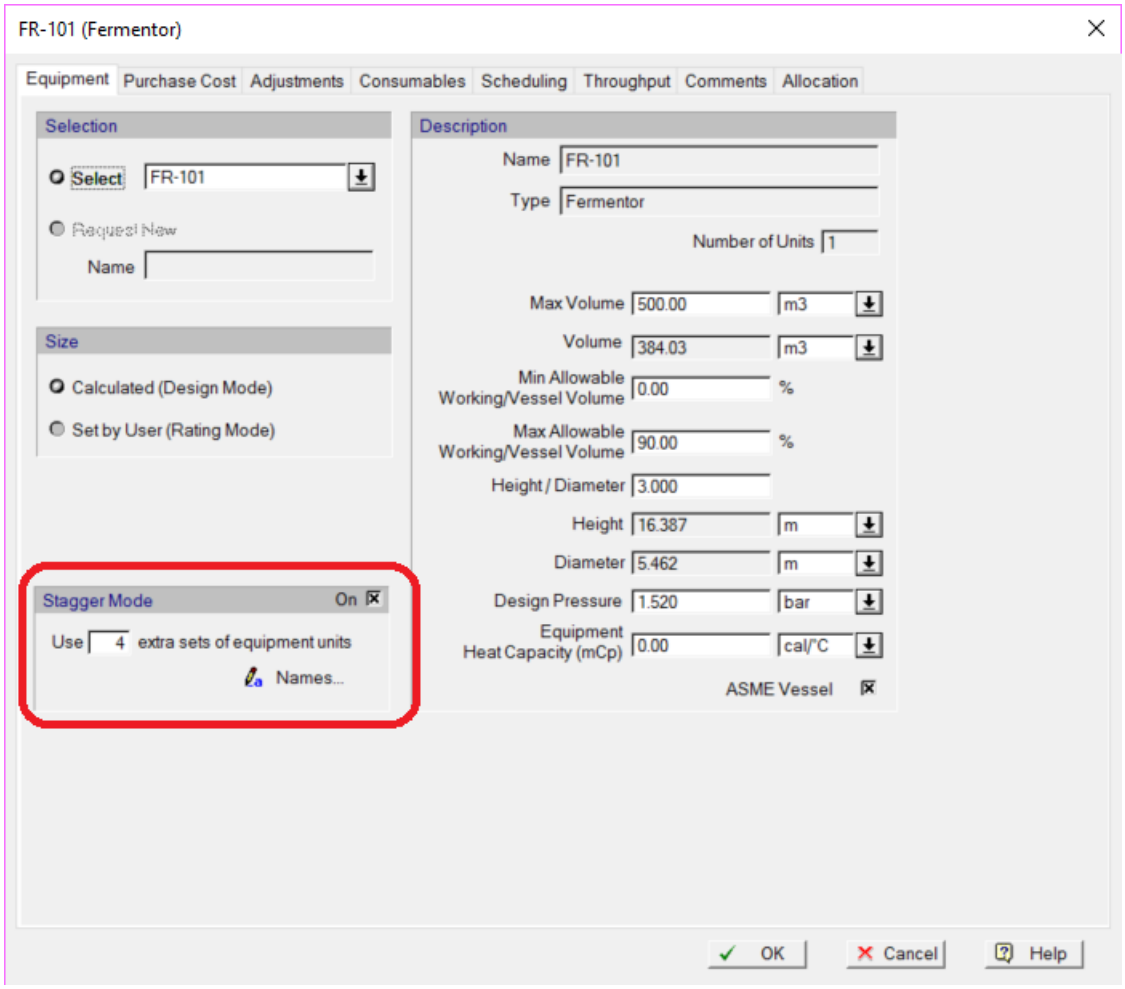


Figure 5 Equipment Data Dialog for Fermentor FR-101.

Storage Units

Storage units (SUs) are virtual units (i.e., they are not displayed on the flowsheet) that allow the user to supply or collect materials needed or generated during a process. For example, supply SUs may be used to provide the cleaning material required by cleaning-in-place (CIP) operations, while receiving SUs may be used to collect the waste from output streams or CIP operations.

Storage units can be added to a SuperPro model by selecting **Tasks \ Other Resources \ Material Storage Units** or by right-clicking on an empty space on the flowsheet and selecting **Resources \ Material Storage Units**. In the dialog window that appears, supply and receiving storage units are displayed on separate tabs. To add a new unit, click on the Add New Receiving (or Supply) Storage Unit button ( ). Selecting a unit and clicking on the spectacles button ( ) brings up a dialog through which you can view and modify the properties of the unit. The properties dialog includes several tabs. The first is

the **Properties** tab through which you can specify the name of the unit, the material supplied or deposited and information related to its cost. The second tab allows you to specify the **Availability Limits** of the storage unit. The third tab is the **Inventory Data** tab which will be discussed in detail below, and the final tab is the **Comments** tab. By clicking the  button, the program will display a dialog indicating the locations (i.e., procedures, operations, and streams) where the storage unit is engaged. The Receiving units tab contains two more buttons, one for displaying the contents of the storage unit () and another for displaying the environmental properties of its contents after a user-specified operation (). The properties of storage units can also be accessed through the "SUs" tab of the **Process Explorer** pane. The Process Explorer pane is displayed by selecting **View \ Process Explorer** or by clicking on the Process Explorer button () of the main toolbar.

To assign a Supply storage unit to an input stream, right-click on the stream and select *Assign Supply Storage Unit* (note that this option is only available in projects which already contain at least one Supply storage unit). The stream composition is automatically set in accordance with the contents of the Supply storage unit. In addition, Supply storage units can be assigned to CIP operations by selecting the appropriate unit in the *Drawn from Storage Unit* dropdown list in the Operating Conditions tab. In this example, a supply storage unit named *NaOH Supply Unit* is used by the streams associated with the Pull-In NaOH operations as well as the post-fermentation "Wash with NaOH" CIP sub-steps within the seed and production fermentors.

To assign a receiving storage unit to an output stream, right-click on the stream and select *Assign Receiving Storage Unit* (note that this option is only available in projects which already contain at least one Receiving storage unit). Receiving storage units can also be assigned to a CIP operation by the Waste *Deposited to Storage Unit* dropdown list in the operating conditions tab. Notice that in this case, the waste is automatically classified according to the type of waste which has been defined for the receiving unit (e.g., aqueous waste, organic waste, etc.). Moreover, if the *Remove Contents as Waste with Disposed Agent* option is checked for a CIP step, any material present in the equipment prior to the CIP operation is removed as CIP waste. In this example, a receiving storage unit named *Wastewater Tank* is used to collect the Aq-Waste stream coming out of the P-36 cooler in the IPA Recovery section as well as the liquid wastes generated by the CIP operations in the fermentors.

Once the material balances have been solved, the program offers the possibility to create storage unit receiving and inventory charts. To generate a receiving unit chart, click **Charts \ Storage Units \ Receiving \ Multiple Batches** and select the storage unit of interest. Figure 6 displays the receiving chart of the **Wastewater Tank** storage unit for 15 consecutive batches. The red lines represent the instantaneous deposit rate and correspond to the y-axis on the left hand side. The green lines represent the daily deposit amounts and correspond to the y-axis on the right hand side.

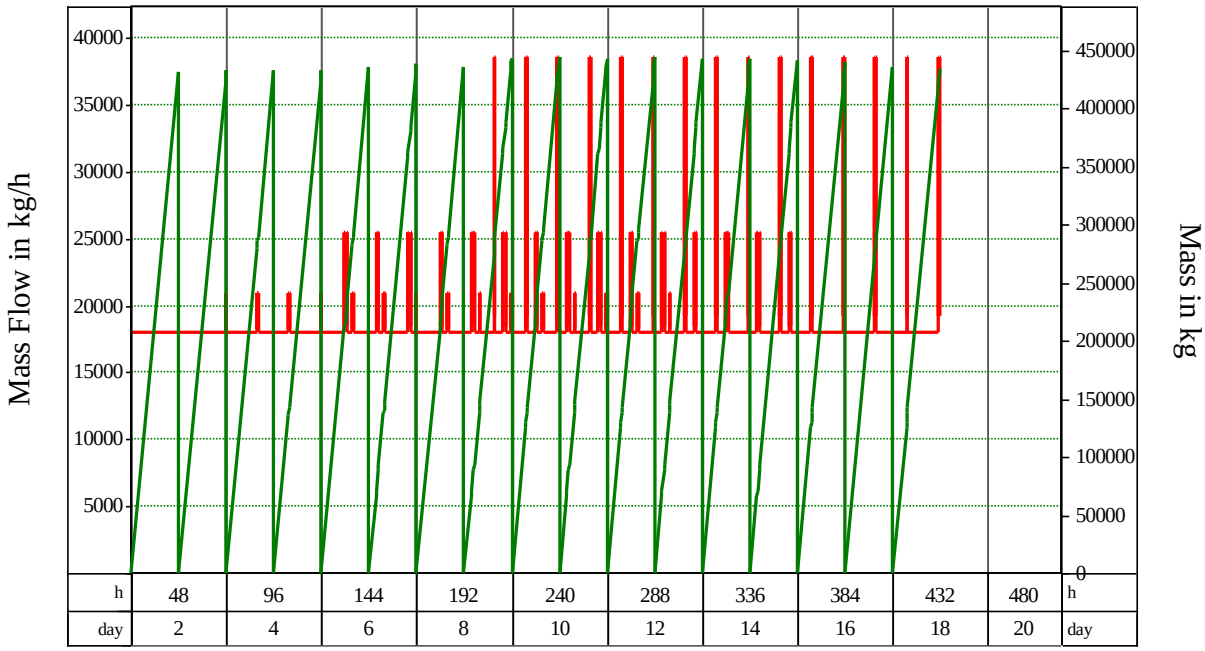


Figure 6 Wastewater Receiving Storage Unit Chart.

Energy Integration

In the xanthan gum process, there are multiple operations that either consume or generate heat. In order to improve the efficiency and economics of the process by reducing the overall utility usage and associated costs, heat can be exchanged between hot streams (heat sources) and cold streams (heat sinks). This can be accomplished in SuperPro Designer by one of the following methods:

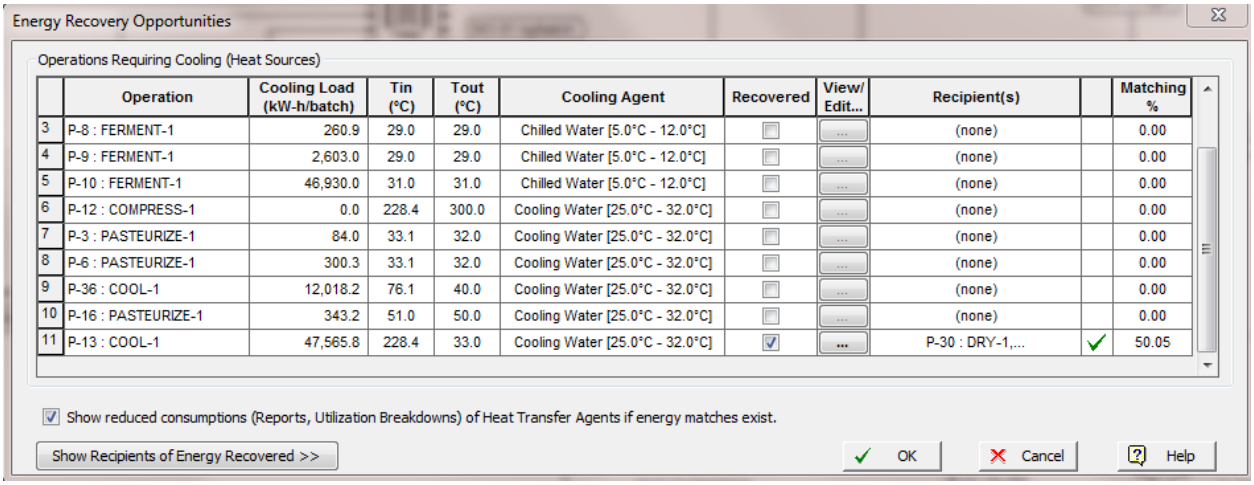
1) You can add heat exchanging procedures directly to the flowsheet by selecting **Unit Procedures \ Heat Exchange \ Heat Exchanging** from the main menu, connecting the relevant input and output streams, and defining the operational data for the operation.
2) You can specify a virtual (i.e., not displayed on the flowsheet) heat recovery network by right-clicking on the flowsheet and selecting **Energy Recovery**.

The first method is used in the IPA recovery section of the process. The distillation feed is preheated, in sequence, against both the distillate and the bottoms streams coming out of the distillation column. As a result, its temperature rises to 80°C, thereby reducing the reboiler thermal requirement of the column. At the same time, the temperature of the aqueous waste stream is reduced from 100°C to 76°C, which results in a decrease of the cooling load for the P-36 cooler.

The second method is used to exploit the extremely large thermal load needed for the cooling of the pressurized fermentation air in P-13 (~2973 kW) in order to fully accommodate the heating requirements of the two dryers of the process, P-21 (~1358 kW) and P-30 (~130 kW). This method allows for a simple

and quick set-up of a heat recovery network that can be easily inspected and modified from a single dialog and eliminates the need for additional stream lines and unit procedure icons on the flowsheet.

To access the Energy Recovery dialog, right-click on a blank area of the flowsheet and select **Energy Recovery** from the menu that appears. The Energy Recovery Opportunities window (see Figure 7) displays all operations in the model that require cooling, the respective cooling loads, as well as their inlet and outlet temperatures. For instance, the COOL-1 operation in procedure P-13 requires 47,565.8 kWh/batch (which is equivalent to 2972 kW because a new batch is initiated every 16 h) in order to reduce the temperature of the fermentation air from 228.4°C to 33°C.



	Operation	Cooling Load (kW-h/batch)	Tin (°C)	Tout (°C)	Cooling Agent	Recovered	View/ Edit...	Recipient(s)		Matching %
3	P-8 : FERMENT-1	260.9	29.0	29.0	Chilled Water [5.0°C - 12.0°C]	☐	...	(none)		0.00
4	P-9 : FERMENT-1	2,603.0	29.0	29.0	Chilled Water [5.0°C - 12.0°C]	☐	...	(none)		0.00
5	P-10 : FERMENT-1	46,930.0	31.0	31.0	Chilled Water [5.0°C - 12.0°C]	☐	...	(none)		0.00
6	P-12 : COMPRESS-1	0.0	228.4	300.0	Cooling Water [25.0°C - 32.0°C]	☐	...	(none)		0.00
7	P-3 : PASTEURIZE-1	84.0	33.1	32.0	Cooling Water [25.0°C - 32.0°C]	☐	...	(none)		0.00
8	P-6 : PASTEURIZE-1	300.3	33.1	32.0	Cooling Water [25.0°C - 32.0°C]	☐	...	(none)		0.00
9	P-36 : COOL-1	12,018.2	76.1	40.0	Cooling Water [25.0°C - 32.0°C]	☐	...	(none)		0.00
10	P-16 : PASTEURIZE-1	343.2	51.0	50.0	Cooling Water [25.0°C - 32.0°C]	☐	...	(none)		0.00
11	P-13 : COOL-1	47,565.8	228.4	33.0	Cooling Water [25.0°C - 32.0°C]	☑	...	P-30 : DRY-1,...	✓	50.05

Figure 7 The Energy Recovery dialog window.

A heat integration network can be specified by matching operations that require cooling with operations that require heating. To display the operations that require heating, click on the **Show Recipients of Energy Recovered >>** button on the bottom left of the dialog. This will expand the Energy Recovery dialog.

As shown in the expanded Energy Recovery dialog (Figure 8), the COOL-1 operation requires an amount of cooling that is sufficient (and at an appropriately high initial temperature) to satisfy the heating loads of the drying operations in both P-21 and P-30. To set up a heat exchange match between these operations, first click on the checkbox in the **Recovered** column for the P-13/COOL-1 operation so that its ellipse button () in the **View/Edit** column is activated. By clicking the ellipse button for the P-13/COOL-1 operation, the **Recovered Energy Matches** dialog window appears (Figure 9).

The Recovered Energy Matches dialog displays the available operations that could potentially act as heat recipients for the P-13/COOL-1 operation. In order to select a heat recipient operation to match with the COOL-1 operation in P-13, ***double-click*** on the respective **Match** check box. Notice that the "Match %" is calculated and displayed on-the-fly in the respective column. As can be seen from Figure 9, the utility requirements for the drying operations in P-21 and P-30 can be satisfied completely, thereby eliminating the hearing requirements of the dryers. In addition, the matching reduces the cooling requirement of P-13 by 50%.

The heat integration savings are reported directly beneath the Utilities Cost table of the Economic Evaluation Report (EER). There it can be seen that the utility reduction from the specified match is approximately $686,000/year.

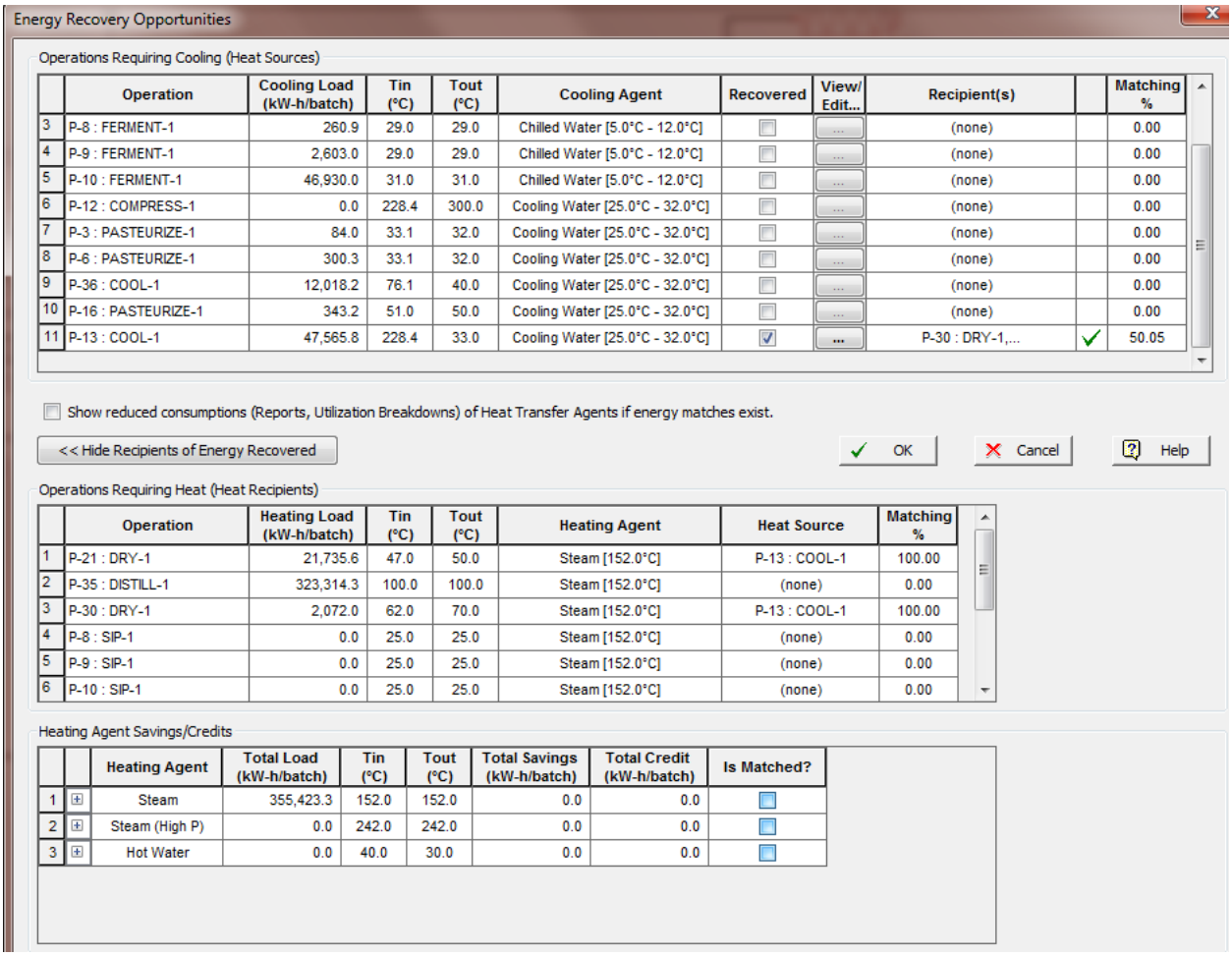
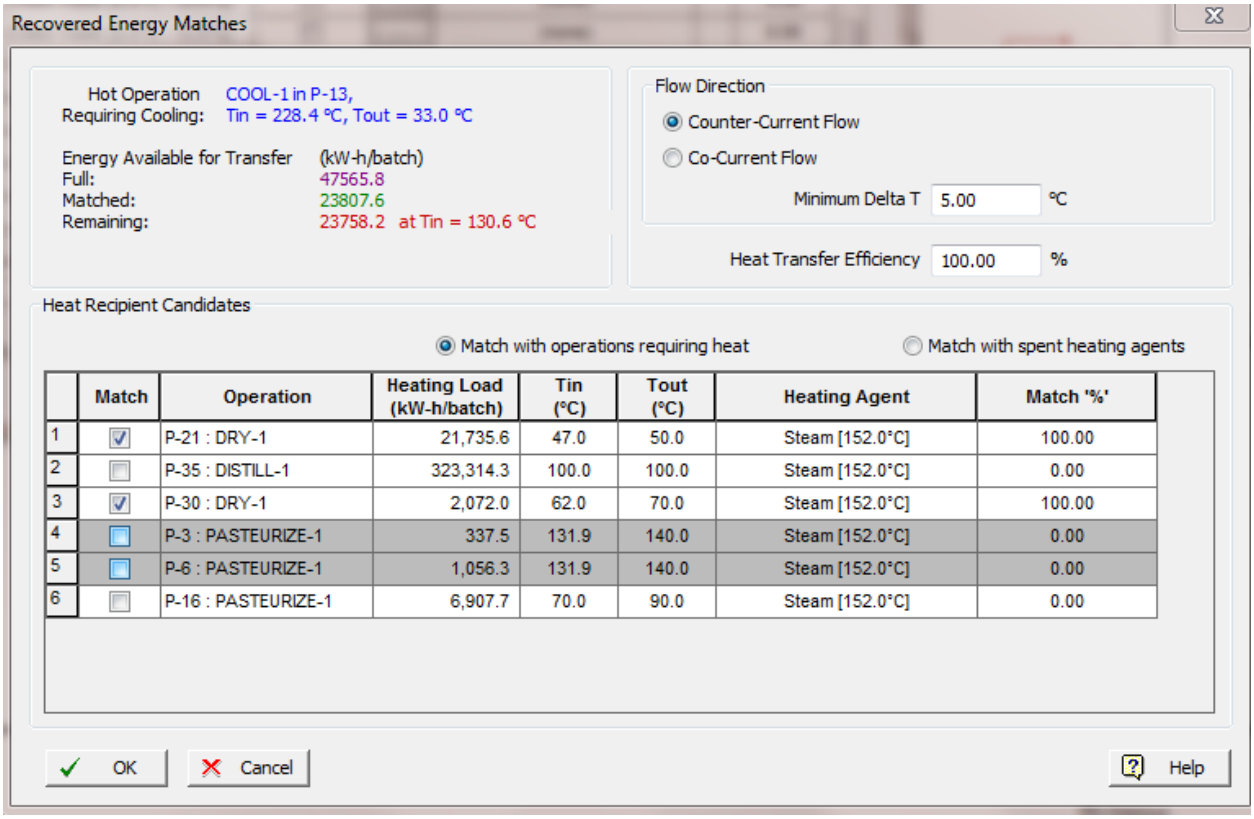


Figure 8 The expanded Energy Recovery dialog window.



	Match	Operation	Heating Load (kW-h/batch)	Tin (°C)	Tout (°C)	Heating Agent	Match '%'
1	☑	P-21 : DRY-1	21,735.6	47.0	50.0	Steam [152.0°C]	100.00
2	☐	P-35 : DISTILL-1	323,314.3	100.0	100.0	Steam [152.0°C]	0.00
3	☑	P-30 : DRY-1	2,072.0	62.0	70.0	Steam [152.0°C]	100.00
4	☐	P-3 : PASTEURIZE-1	337.5	131.9	140.0	Steam [152.0°C]	0.00
5	☐	P-6 : PASTEURIZE-1	1,056.3	131.9	140.0	Steam [152.0°C]	0.00
6	☐	P-16 : PASTEURIZE-1	6,907.7	70.0	90.0	Steam [152.0°C]	0.00

Figure 9 The Recovered Energy Matches dialog window.

Economic Evaluation

This plant produces 620 kg/h of xanthan gum during an operating period of 8208 hours (342 days) per year, which corresponds to around 5000 MT/year. Various assumptions were made for the costs of raw materials, heat transfer agents, wastewater treatment, equipment purchase costs, labor, *etc*. In addition to the mass and energy balances, SuperPro Designer calculates the capital investment (CAPEX), operating expenses (OPEX), the production cost, and the profitability of the project. These results are reported in the Economic Evaluation (EER), Cash Flow Analysis (CFR), Itemized Cost (ICR), and Excel Custom reports.

The information on equipment sizes and purchase costs for this process is displayed in Table 1 (extracted from the EER). **User-Defined Equipment Cost Models** were specified for the distillation column, the fermentors, the buffer tanks, and the compressor. Detailed information on how to specify **User-Defined Equipment Cost Models** can be found in the corresponding section of the Corn Refinery example's ReadMe file.

Table 1 Major equipment specification and FOB cost (2016 prices).

Quantity/ Standby/	Name	Description	Unit Cost ($)	Cost ($)
1 / 0 / 0	V-104	Blending Tank	167,000	167,000
		Vessel Volume = 341.38 m3		
1 / 0 / 0	DC-101	Decanter Centrifuge	258,000	258,000
		Throughput = 19332.43 L/h		
1 / 0 / 0	DC-102	Decanter Centrifuge	258,000	258,000
		Throughput = 3998.93 L/h		
1 / 0 / 0	RDR-101	Rotary Dryer	663,000	663,000
		Drying Area = 114.19 m2		
1 / 0 / 0	PFR-101	Plug Flow Reactor	239,000	239,000
		Vessel Volume = 229.28 m3		
1 / 0 / 0	DC-103	Decanter Centrifuge	258,000	258,000
		Throughput = 206358.80 L/h		
1 / 0 / 0	BF-101	Belt Filter	279,000	279,000
		Belt Width = 1.82 m		
1 / 0 / 0	HX-101	Condenser	34,000	34,000
		Condensation Area = 7.15 m2		
1 / 0 / 0	HX-102	Heat Exchanger	11,000	11,000
		Heat Exchange Area = 1.96 m2		
1 / 0 / 0	HX-103	Heat Exchanger	13,000	13,000
		Heat Exchange Area = 2.59 m2		
1 / 0 / 0	C-101	Distillation Column	1,000,000	1,000,000
		Column Volume = 7.55 m3		
1 / 0 / 0	DDR-101	Drum Dryer	179,000	179,000
		Drum Area = 23.65 m2		
1 / 0 / 2	SFR-101	Seed Fermentor	18,000	54,000
		Vessel Volume = 1.13 m3		
1 / 0 / 3	SFR-201	Seed Fermentor	97,000	388,000

		Vessel Volume = 18.80 m3		
1 / 0 / 4	FR-101	Fermentor	592,000	2,960,000
		Vessel Volume = 384.03 m3		
1 / 0 / 0	G-101	Centrifugal Compressor	1,035,000	1,035,000
		Compressor Power = 3076.25 kW		
1 / 0 / 0	V-101	Blending Tank	36,000	36,000
		Vessel Volume = 26.33 m3		
1 / 0 / 0	PZ-101	Pasteurizer	401,000	401,000
		Rated Throughput = 4739.93 L/h		
1 / 0 / 0	V-103	Blending Tank	28,000	28,000
		Vessel Volume = 17.20 m3		
1 / 0 / 0	PZ-102	Pasteurizer	910,000	910,000
		Rated Throughput = 15477.00 L/h		
1 / 0 / 0	V-102	Flat Bottom Tank	72,000	72,000
		Vessel Volume = 84.27 m3		
1 / 0 / 0	HX-104	Heat Exchanger	44,000	44,000
		Heat Exchange Area = 18.57 m2		
1 / 0 / 0	PZ-103	Pasteurizer	127,000	127,000
		Rated Throughput = 19202.70 L/h		
1 / 0 / 0	AF-101	Air Filter	172,000	172,000
		Rated Throughput = 45407.35 m3/h		
1 / 0 / 0	HX-101	Heat Exchanger	63,000	63,000
		Heat Exchange Area = 21.51 m2		
		Unlisted Equipment		1,986,000
			TOTAL	11,768,000

Table 2 (also extracted from the EER) provides an estimate of the Fixed Capital Estimate Summary, which is around $58.8 million.

Table 2 Fixed capital investment summary (2016 prices in $).

2A. Total Plant Direct Cost (TPDC) (physical cost)	
1. Equipment Purchase Cost	11,768,000
2. Installation	4,300,000
3. Process Piping	3,015,000
4. Instrumentation	3,236,000
5. Insulation	353,000
6. Electrical	1,177,000
7. Buildings	3,457,000
8. Yard Improvement	1,397,000
9. Auxiliary Facilities	3,236,000
TPDC	31,939,000

2B. Total Plant Indirect Cost (TPIC)	
10. Engineering	7,985,000
11. Construction	11,179,000
TPIC	19,163,000

2C. Total Plant Cost (TPC = TPDC+TPIC)	
TPC	51,102,000

2D. Contractor's Fee & Contingency (CFC)	
12. Contractor's Fee	2,555,000
13. Contingency	5,110,000
CFC = 12+13	7,665,000

2E. Direct Fixed Capital Cost (DFC = TPC+CFC)	
DFC	58,768,000

Table 3 displays information on the cost of raw materials. Glucose is the dominant raw material cost. A price of $0.40/kg was assumed for glucose. The total annual cost of raw materials is around $4.2 million. The cost of isopropanol (IPA) is small because the vast majority of IPA is purified and reused in the process.

Table 3 Materials Cost – Process Summary.

Bulk Material	Unit Cost ($)	Annual Amount		Annual Cost ($)	%
Air	0.00	616,319,102	kg	0	0.00
Amm. Nitrate	0.15	184,533	kg	27,680	0.66
Biomass	0.00	841	kg	0	0.00
Citric Acid	0.50	337,683	kg	168,841	4.03
Glucose	0.40	8,017,777	kg	3,207,111	76.48
Isopropanol	0.60	50,273	kg	30,164	0.72
KH2PO4	0.30	482,475	kg	144,742	3.45
NaOH (20% w/w)	0.01	586,271	kg	4,655	0.11
Salts	0.50	215,439	kg	107,720	2.57
Water	3.00	167,502	MT	502,505	11.98
Xanthan	0.00	153	kg	0	0.00
TOTAL				4,193,419	100.00

NOTE: Bulk material consumption amount includes material used as:
- Raw Material
- Cleaning Agent
- Heat Transfer Agent (if utilities are included in the operating cost)

Table 4 displays the utilities costs. Notice that the steam demand of the process is about 61 MT per MT of xanthan gum product. The annual expense for utilities is around $16 million, which is roughly four times the cost of materials. Table 5 displays the cost of labor, which is around $6 million per year.

Table 4 Utilities Cost.

Utility	Unit Cost ($)	Annual Amount	Ref. Units	Annual Cost ($)	%
Std Power	0.10	44,220,023	kW-h	4,422,002	27.59
Steam	30.00	305,266	MT	9,157,981	57.15
Cooling Water	0.05	23,528,356	MT	1,176,418	7.34
Chilled Water	0.40	3,172,914	MT	1,269,166	7.92
TOTAL				16,025,567	100.00

Table 5 Labor Cost – Process Summary.

Labor Type	Unit Cost ($/h)	Annual Amount	Annual Cost ($)	%
Operator	69.00	85,125	5,873,621	100.00
TOTAL		85,125	5,873,621	100.00

Figure 10 displays the breakdown of the annual operating cost. It can be seen that the utilities, followed by the facility-dependent costs (annualized fixed capital investment, maintenance, etc.) have the highest contribution to the total annual operating costs.

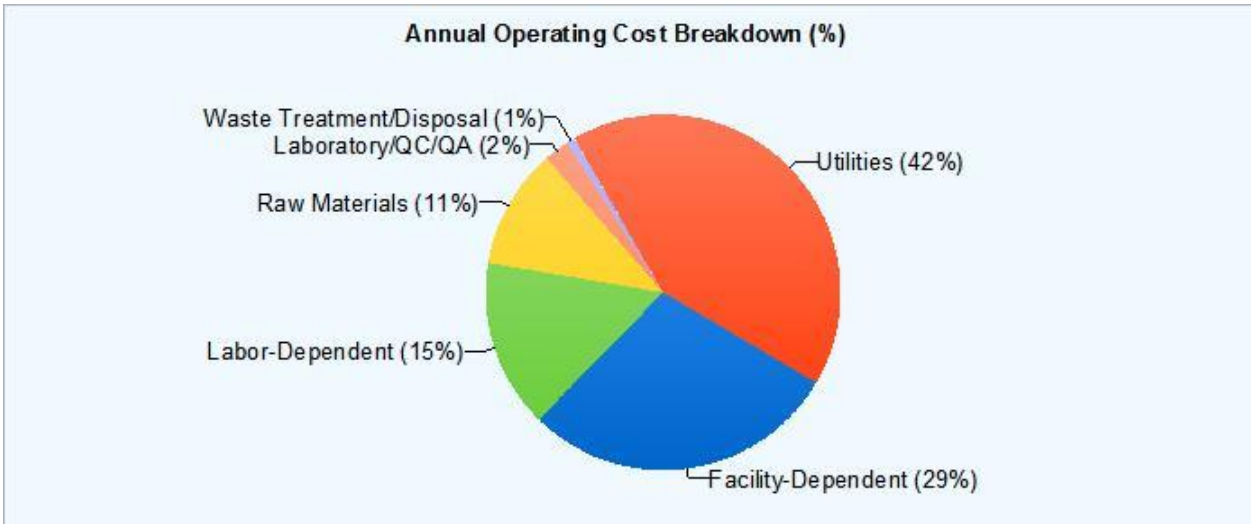


Figure 10 Annual Operating Cost Breakdown (%).

The executive summary of the project is presented in Table 6. The total CAPEX for such a plant is around $64 million. The annual operating cost is around $38.5 million, which corresponds to a unit production cost of around $7.56/kg. This is a realistic figure for a non-depreciated facility. Assuming a selling price of $9/kg for xanthan gum, the expected payback time is around 6.4 years.

Table 6 Executive Summary (2016 prices).

Total Capital Investment	64,023,000	$
Capital Investment Charged to This Project	64,023,000	$
Operating Cost	38,533,000	$/yr
Savings (due to Heat Recovery)	868,491	$/yr
Main Revenue	44,851,000	$/yr

Other Revenues	102,678	$/yr
Total Revenues	44,954,000	$/yr
Cost Basis Annual Rate	4,983,474	kg MP/yr
Unit Production Cost	7.73	$/kg MP
Net Unit Production Cost	7.56	$/kg MP
Unit Production Revenue	9.02	$/kg MP
Gross Margin	16.22	%
Return On Investment	15.55	%
Payback Time	6.43	years
IRR (After Taxes)	7.73	%
NPV (at 7.0% Interest)	3,125,000	$

MP = Total Flow of Stream 'Xanthan Gum Product'

Summary

Our objective in this example was to present a simple xanthan gum process model in SuperPro Designer, which can be used as a reference for modeling similar types of processes. Key results based on the assumptions in this model include the determination that a manufacturing plant with a capacity of 5000 MT/year requires a total CAPEX of around $64 million and annual operating expenditures (including depreciation) of around $38.5 million. Moreover, for every kg of xanthan gum produced, the process requires around 33.5 kg of water, 61 kg of steam, 29 kg of IPA and 5.3 m^3 of circulated cooling and chilling water.

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Xanthan Gum Production - Precipitation with IPA

